

Jingtao Zhang

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INFORMATION

EDUCATION **Virginia Tech**, Blacksburg, USA

Ph.D., Operations Research, 8/16/2021–8/4/2025

- Courses taken: Optimization, Machine Learning for System Intelligence, Dynamic Programming, Advanced Mathematical Programming, Random Process, Measure and Probability, Statistical Inference, Surrogate Modeling

Shanghai Jiao Tong University, Shanghai, China

M.E., Logistics Engineering, 9/1/2017–3/30/2020

- Courses taken: Stochastic Models, Transportation and Distribution, Sustainable Supply Chain Management, Production and Operations Management
- GPA Rank 1/13

Dalian Maritime University, Liaoning, China

B.E., Logistics Engineering, 9/1/2013–6/30/2017

- Courses taken: Logistics Information Management, Systems Engineering, Techniques and Method of Forecasting, Simulation Technology of Logistics System
- GPA Rank 1/67

RESEARCH Design and analysis of stochastic simulation experiment, uncertainty quantification,
INTERESTS simulation optimization, active learning, sensitivity analysis

WORKING **Research Scientist Intern at Amazon** 5/20/2024 – 8/23/2024
EXPERIENCE

Supply Chain Optimization Technologies (SCOT)

Job description: I developed an uncertainty quantification framework to predict quantiles and construct prediction intervals for critical supply chain metrics, achieving a nearly 10% improvement in prediction accuracy. Additionally, I enhanced the framework to enable prediction attribution using Shapley values.

Research Scientist Intern at Amazon 5/31/2023 – 8/18/2023

Supply Chain Optimization Technologies (SCOT)

Job description: I developed an advanced simulation module for modeling order quantities in the supply chain, delivering significantly improved accuracy and substantially faster simulation speeds compared to the existing production system. The module was implemented and packaged as a core component in Python.

RESEARCH EXPERIENCE	Nested simulation for global sensitivity analysis	Dec. 2023 – Present
	<i>Grado Department of Industrial and Systems Engineering, Virginia Tech</i>	
	<i>Advisor: Prof. Xi Chen</i>	
	Active learning for robust optimization via metamodeling	July. 2023 – Present
	<i>Grado Department of Industrial and Systems Engineering, Virginia Tech</i>	
	<i>Advisor: Prof. Xi Chen</i>	
	Asymptotic normality and efficiency of dual metamodel-based Sobol' index estimators	June. 2022 – Mar. 2023
	<i>Grado Department Of Industrial And Systems Engineering, Virginia Tech</i>	
	<i>Advisor: Prof. Xi Chen</i>	
	Multilevel Monte Carlo (MLMC) metamodeling for variance function estimation	July. 2021 – Feb. 2023
	<i>Grado Department Of Industrial And Systems Engineering, Virginia Tech</i>	
	<i>Advisor: Prof. Xi Chen</i>	
	Multi-stage lot-sizing and scheduling in automobile powertrain manufacturing process	July. 2019 – Feb. 2020
	<i>Sino-US Global Logistics Institute, Shanghai Jiao Tong University</i>	
	<i>Advisor: Prof. Jianghang Chen</i>	
	Lot-sizing and scheduling with continuous demand in automobile powertrain manufacturing process	Feb. 2018 – July. 2019
	<i>Sino-US Global Logistics Institute, Shanghai Jiao Tong University</i>	
	<i>Advisor: Prof. Jianghang Chen</i>	
	Integrated continuous berth allocation and quay crane scheduling under tidal influence	Nov. 2016 – July. 2017
	<i>Transportation Management College, Dalian Maritime University</i>	
	<i>Advisor: Prof. Hongxing Zheng</i>	
PUBLICATION	• J. Zhang, X. Chen, “Sobol' Index Estimation via Nested Simulation,” to be submitted. • J. Zhang, X. Chen, “A Multilevel Monte Carlo Metamodeling Approach to Variance Function Estimation,” <i>SIAM Journal on Uncertainty Quantification</i>. • X. Chen, Y. Zhang, G. Xie, J. Zhang, “A Uniform Error Bound for Stochastic Kriging: Properties and Implications on Simulation Experimental Design,” <i>ACM Transactions on Modeling and Computer Simulation</i>, 2024. • J. Zhang, X. Chen, R. Wang, “Asymptotic Normality of Joint Metamodel-based Sobol' Index Estimators,” <i>2023 Winter Simulation Conference</i>, 3705–3716, 2023. • J. Zhang, J. Chen, “Optimization of Multiple Distribution Center Location Problem Considering Flow Balance,” <i>Logistics Sci-Tech</i>, 42(10):15-17,	

2019.

- H. Zheng, J. Zhang, B. Liu, “Integrated continuous berth allocation and quay crane scheduling under tidal influence at container terminal,” *Computer Integrated Manufacturing Systems*, 24(10):2599-2611, 2018.

PROJECTS

Multiple distribution center location for SAIC General Motors Company 2019

Choose several distribution center locations using an improved K-means method

Intelligent powertrain production planning system in SAIC General Motors Company 2017 – 2018

Build a digital production planning system for an automobile company

Intelligent analysis method of traffic saturation in ship alignment system 2014

Design a method for measuring the traffic saturation in specific waters

AWARDS

Virginia Tech

- Graduate fellowship 2021

Shanghai Jiao Tong University

- First-class academic scholarship 2017 – 2020
- Silver award for the 3rd national logistics innovation competition 2018
- Outstanding graduates of Shanghai Jiao Tong University 2019

Dalian Maritime University

- National Merit Scholarship of China (top 1% GPA) 2014 – 2017

SKILLS

- Python, MATLAB, Git, L^AT_EX, Microsoft Office Suite, Elementary Java/Shell/SQL